

From Laboratory to Field: A Leakage-Controlled Cross-Domain Evaluation Protocol for Plant Disease Recognition

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Abstract. Plant disease classifiers trained on laboratory imagery report accuracies above 0.99, yet field deployment disappoints. The size of that gap is widely asserted and rarely measured under conditions that make the measurement trustworthy. We report a cross-domain evaluation in which the two obvious confounds are closed rather than assumed away. First, train/test contamination: an exhaustive perceptual-hash search over every distinct-hash pair across the two corpora, followed by human adjudication of all 16 near candidates, establishes 0 exact overlaps between the laboratory training corpus and the field evaluation corpus. Second, label-space construction: the shared 21-class space is assembled only from a deterministic normaliser applied identically to both label sets and from a previously recorded human policy decision, so no new semantic judgement is introduced to make the comparison look better or worse. Three ImageNet-pretrained backbones are trained under one protocol on 54 305 laboratory images and evaluated on 1 951 field images, with paired significance tests and bias-corrected accelerated bootstrap intervals. Every input is bound to a digest-verified dataset lock (83449f6013691855). We find that near-saturated in-domain accuracy carries almost no information about which backbone transfers, and that a temperature fitted on in-domain validation data—standard practice for calibration—makes out-of-domain calibration substantially worse rather than better.

Keywords: plant disease recognition · cross-domain evaluation · leakage audit · model calibration · reproducibility

1 Introduction

Image-based plant disease recognition has an unusually clean success story and an unusually persistent complaint. The success story is PlantVillage [14]: tens of thousands of labelled leaf photographs on which convolutional networks reach accuracies above 0.99, a result reproduced so widely that the benchmark is effectively saturated. The complaint is that these accuracies do not survive contact with a real field. PlantVillage images are single detached leaves photographed against uniform backgrounds under controlled lighting. A photograph taken by a grower contains occlusion, soil, multiple leaves, variable illumination, and a camera held at whatever distance was convenient.

That a gap exists is not in dispute, and field-condition corpora have been built precisely to expose it [20,15,23]. What is harder to find is a measurement of the gap that a sceptical reader can trust, because two confounds sit between the raw comparison and any conclusion drawn from it.

The first is contamination. If any field evaluation image is a duplicate or near-duplicate of a laboratory training image, the measured transfer is optimistic by an unknown amount. This is not hypothetical: near-duplicate contamination has been documented in benchmarks far more scrutinised than these [1]. Most cross-corpus plant disease comparisons do not report a leakage audit at all.

The second is the label space. The two corpora do not share a taxonomy, so any cross-corpus evaluation must construct a correspondence between them. That construction is a research degree of freedom. A generous mapping that pairs biologically similar classes will report a smaller gap than a conservative one, and the reader usually cannot tell which was used or what was excluded.

This paper closes both. Our contributions are:

1. **A leakage-controlled evaluation protocol.** Candidate duplicate pairs are generated by exhaustive perceptual-hash comparison over every distinct-hash pair across the two corpora—not by an approximate index—and every candidate is adjudicated by a human. Perceptual hashing generates candidates; it never decides. The result is 0 exact and 16 adjudicated-distinct near pairs between the corpora, so the measured degradation is a generalisation measurement rather than an artefact.
2. **An evidence-based shared label space.** The 21-class cross-domain space is assembled from exactly two sources: a deterministic normaliser applied unchanged to both label sets, and a human policy decision recorded before this evaluation was designed. Every excluded class carries a written reason, including biologically plausible near-misses that we decline to assert. The mapping is conservative by construction and we say what a more generous one would change.
3. **A controlled multi-seed comparison with paired statistics.** Three backbones spanning a $15\times$ parameter range are trained under one protocol across independent seeds, evaluated in-domain and cross-domain, and compared with McNemar’s exact test on per-item correctness [13,4] and bias-corrected accelerated bootstrap intervals [5], with family-wise error control [11].
4. **Two findings that matter for practice.** In-domain ranking on a saturated benchmark does not predict cross-domain ranking; and a temperature fitted on in-domain validation data, applied across the domain shift, degrades calibration rather than repairing it—the opposite of the standard recommendation.

Every input is bound to a digest-verified dataset lock, every reported number is generated from run artefacts by script, and no number in this document was typed by hand.

2 Related Work

Laboratory corpora and their saturation. PlantVillage [14] established the standard benchmark and, with it, the standard result: accuracies above 0.99 across dozens of crop–disease classes. The corpus’s construction—detached leaves, uniform backgrounds—is stated plainly in the original work. What followed was a long sequence of papers reporting incremental gains on a benchmark with very little headroom, a pattern that makes in-domain comparison progressively less informative.

Field-condition corpora. PlantDoc [20] was assembled specifically to test whether laboratory-trained models transfer, and reported that they largely do not. FieldPlant [15] and PlantWild [23] extend the field-condition setting to larger and more varied imagery, and PlantSeg [24] adds segmentation supervision. Our work is complementary: rather than proposing a new corpus or a stronger model, we ask what conditions a cross-corpus comparison must satisfy before its number means anything.

Generalisation and dataset bias. The observation that models absorb corpus-specific regularities rather than the intended concept predates deep learning [22] and has been quantified repeatedly since [18,8]. For plant disease, the uniform background of laboratory imagery is an obvious shortcut candidate: it is perfectly correlated with the corpus and trivially easy to represent. Near-duplicate leakage is the adjacent failure, and [1] showed it survives in benchmarks that had been studied for a decade. We treat both as things to measure, not to assume.

Calibration and selective prediction. Modern networks are systematically over-confident, and temperature scaling is the standard single-parameter remedy [9]. Under distribution shift, uncertainty estimates degrade further and the ranking of calibration methods changes [17]. Selective prediction [7] offers the deployment-relevant view: not how accurate a model is, but how accurate it is on the subset it is willing to act on. All three matter for an agricultural decision system, where an unreliable confident prediction can cost a season.

Positioning within this venue. LeADS [2] carries a leaf-image pipeline through stress and disease detection to severity estimation, establishing the applied trajectory this community is on. AgroBench [19] shows how far general-purpose vision–language models remain from reliable agricultural judgement. Our contribution sits underneath both: whatever the downstream stage, its evaluation inherits whatever contamination and label-space slack exist in the cross-domain measurement it rests on. We aim to make that foundation auditable.

3 Datasets and Leakage Control

3.1 Corpora

PlantVillage supplies the laboratory domain: 54 305 colour images over 38 crop-disease classes, split 43 596 train / 10 709 test. The corpus is pinned to an immutable upstream revision and the manifest is reconstructed exactly from the pinned split files, with every image verified against its recorded digest.

PlantDoc [20] supplies the field domain. 2 578 records were acquired at a pinned source revision. Two reductions follow, both recorded with reasons: 14 images are removed by human adjudication of byte-exact duplicate groups, and 3 further images are *excluded, not relabelled*, because the diagnostic evidence for a relabel was judged insufficient and the second scientific review that would settle it has not been performed. The exclusion records no diagnosis and asserts nothing about what those images depict. The resulting *PlantDoc Core* holds 2 561 images over 28 classes, split 2 336 train / 225 test. No source image was deleted and no image byte was modified.

Class support in PlantDoc Core is severely uneven—the largest training class holds ninety times the smallest—and one arthropod-pest class has two training images and no test images at all. It cannot be learned or measured, and is retained only because removing it would alter the locked corpus. Table 1 gives the composition; the full provenance and leakage counts are in the supplementary material.

Table 1. Dataset composition after leakage control and conservative exclusion. The cross-domain evaluation subset holds 1951 PlantDoc Core images over 21 shared classes.

Dataset	Images	Train	Test	Classes
PlantVillage	54305	43596	10709	38
PlantDoc Core	2561	2336	225	28

3.2 Cross-Corpus Leakage

If a field evaluation image duplicates a laboratory training image, the cross-domain number is contaminated. We therefore search for such pairs exhaustively.

Candidate generation uses a 64-bit perceptual hash with exact chunked brute-force Hamming evaluation over *every* distinct-hash pair, which for the acquired population is 1.37×10^8 distance evaluations. The search is exhaustive by construction rather than index-approximate: an approximate nearest-neighbour index would make the absence of a match an artefact of the index’s recall, which is exactly the property this audit must not have. Pairs at Hamming distance ≤ 6 are surfaced as candidates.

Perceptual hashing generates candidates; it never decides. All 16 surfaced near pairs were adjudicated by a human as clearly different, and the great majority are cross-class collisions on plain green foliage—the failure mode a perceptual hash is expected to have. The audit finds 0 exact overlaps and 0 unresolved pairs. No laboratory training image is a byte- or pixel-duplicate of any field evaluation image, so the cross-domain measurement in Section 6 is a genuine generalisation measurement.

Within-corpus leakage is controlled separately. PlantVillage records which physical leaf each image photographs, and several images can show one leaf. The upstream train/test split is leaf-grouped; the validation split we carve from train is *also* leaf-grouped, so no physical leaf appears on both sides of model selection. Ignoring this would let a model be selected on a near-copy of something it trained on.

4 Experimental Protocol

4.1 The Shared Label Space

The two corpora use different taxonomies, so cross-domain evaluation requires a correspondence. Because that correspondence is a research degree of freedom, we constrain it to evidence that existed before this evaluation was designed, from exactly two sources.

Source 1: deterministic key collision. The project’s crop and disease normalisation functions are applied *unchanged and identically* to both label sets. Eleven pairs collide on the resulting canonical key. The same function runs on both sides, so no human judgement enters the pairing.

Source 2: a recorded human policy. A previously approved policy assigns every PlantDoc class of the form “*Crop* leaf” a canonical disease of *healthy*; on the PlantVillage side the token is literal in the label. Ten crops appear on both sides under this policy.

The result is 21 shared canonical classes covering 1 951 PlantDoc Core images. Everything else is excluded *with a written reason*. The excluded set deliberately includes biologically plausible near-misses that the normaliser does not align—for instance a gray-leaf-spot class against a *Cercospora* gray-leaf-spot class, and two differently-named pepper leaf-spot classes. Asserting those pairs would be a new semantic judgement made by the authors of the evaluation, in a direction that would improve the headline number. We decline, and record what a more generous mapping would change. The mapping is conservative by construction, and Section 9 states the consequence.

4.2 Cross-Domain Evaluation Procedure

A laboratory-trained model predicts over its own 38-way space. To evaluate it on the shared space, source-class probabilities are summed within each canonical class and renormalised over the 21 shared classes only. Field images outside the

mapping are *excluded from the evaluation set* rather than counted as errors, since a model cannot be faulted for a class the protocol never asked it about.

Renormalisation discards probability mass, and how much is discarded is itself informative, so the mean retained mass before renormalisation is recorded for every run and reported in Section 6.

All 1 951 shared-class Core images are used, including those PlantDoc assigns to its training split, because a laboratory-trained model has never seen any of them. The split composition is recorded in every result file.

4.3 Models and Training

Three ImageNet-pretrained [3] backbones span an order of magnitude in capacity: ResNet-50 [10] (23.6 M parameters), EfficientNet-B0 [21] (4.0 M), and MobileNetV3-Small [12] (1.5 M). Only the final classifier layer is replaced. Backbone choice is the sole architectural variable.

One protocol governs every run: 224×224 inputs, AdamW at learning rate 3×10^{-4} with weight decay 10^{-4} , cosine decay stepped per batch, mixed precision, batch size 64, label smoothing 0.1, at most 15 epochs with early stopping on validation macro-F1 at patience 3, and selection of the single best-validation checkpoint. Training augmentation is deliberately mild: random resized crop, horizontal flip, rotation within $\pm 15^\circ$, and colour jitter with *near-zero hue* perturbation. Chlorosis and necrosis are diagnosed by colour; a hue shift would destroy the signal the task depends on.

Two deviations from the common protocol are dataset-driven rather than architectural, and both are reported rather than tuned. PlantVillage validation is grouped by leaf identity, as described above. PlantDoc Core training uses inverse-frequency class weighting normalised to mean one, so the effective learning rate is unchanged; with 90:1 imbalance, a class-balanced sampler would replay the two-image arthropod-pest class roughly ninety times per epoch and overfit it, and weighted loss is the gentler instrument.

4.4 Metrics

Classification is reported as accuracy, macro-F1, weighted-F1, balanced accuracy, and macro precision and recall. Macro averages run over the classes *present in the evaluation split*, with the full declared label space reported alongside: averaging in a zero-support class as a hard 0.0 would describe the split rather than the model. Predictions landing on zero-support classes are still errors and are counted explicitly.

Probabilistic quality is reported as negative log-likelihood, Brier score, and expected and maximum calibration error over 15 equal-width bins [16]. The scalar calibration error and the reliability diagram are computed from the same bin table, so they cannot disagree. Selective prediction [7] is reported as a confidence–abstention curve and the area under the risk–coverage curve.

Calibration uses a single temperature fitted by L-BFGS on validation negative log-likelihood and applied unchanged to test and cross-domain logits. It is

never refitted on evaluation data. Temperature does not move the argmax, so accuracy and F1 are identical before and after; only confidence moves. Both the scaled and unscaled variants are reported, which is what makes it possible to ask whether an in-domain calibration fix survives a domain shift.

4.5 Statistical Methodology

Two models evaluated on one dataset are not independent samples, so accuracies are compared with **McNemar’s exact test** on per-item correctness [13,4]. The exact binomial form is used rather than the chi-square approximation because the discordant counts here are small. Because we report every pairwise comparison, p -values are additionally reported with Holm step-down adjustment [11] controlling the family-wise error rate across the whole family of tests.

Interval estimates use the **bias-corrected and accelerated (BCa) bootstrap** [5,6] at the 95% level with 10 000 resamples. Both the bias correction and the acceleration are computed; where either is undefined—as it is for a statistic that does not vary across resamples—the interval degrades to a percentile interval and is labelled as such rather than silently reported as BCa. The evaluated label space is held fixed at the classes present in the original evaluation, so the statistic keeps one definition across all replicates.

These two instruments answer different questions and are reported separately. The bootstrap interval describes sampling variability of a fixed evaluation set for one trained checkpoint; the across-seed standard deviation describes variability of the training process itself. Neither substitutes for the other, and a difference smaller than the seed spread is not claimed as a finding regardless of its p -value.

4.6 Reproducibility

Every run revalidates the dataset lock (83449f60136918554904cb9c596122f8b89cb51ff5bb5c52c541394a5698082d) before training begins, so a checkpoint can always be traced to an exact, digest-bound corpus. Runs are sequential on a single Apple-silicon host with a unified-memory accelerator; timings are therefore Apple-silicon numbers and will not transfer to server GPUs, though relative ordering between backbones should. Seeds are set for Python, NumPy, and the framework at the start of every run, and the validation carve-out depends only on the frame, the fraction, and the seed—not on row order or platform.

Draft status. This version reports 9 of 18 planned runs, covering seeds 42, 1337. All statistical tests in Section 4.5 are computed on seed 42. Sections 5 through 7 are completed once the full matrix finishes; no value in this document is extrapolated from a run that has not happened.

5 In-Domain Results

[Draft: narrative pending completion of the three-seed matrix. The generated tables below are current for the runs that have finished.]

Table 2. In-domain test performance, mean $\pm$ sample standard deviation over three seeds. Weighted-F1, balanced accuracy, and macro precision and recall are in the supplementary table.

Model	Corpus	N	Accuracy	Macro-F1
ResNet-50	PV	10709	0.9925	0.9899
EfficientNet-B0	PV	10709	0.9968	0.9956
MobileNetV3-Small	PV	10709	0.9950 ± 0.0001	0.9929 ± 0.0002
ResNet-50	PDC	225	0.6933	0.6994
EfficientNet-B0	PDC	225	0.6689 ± 0.0157	0.6795 ± 0.0187
MobileNetV3-Small	PDC	225	0.5578 ± 0.0031	0.5893 ± 0.0075

6 Cross-Domain Generalisation

[Draft: narrative pending completion of the three-seed matrix.]

Table 3. Cross-domain generalisation on the frozen shared-class PlantDoc Core subset, mean $\pm$ sample standard deviation over three seeds. Metrics cover only the 21 shared classes; unmapped images are excluded rather than counted as errors. ‘Ret. mass’ is the mean source-model probability retained inside the shared space before renormalisation.

Model	N	Accuracy	Macro-F1	Ret. mass
ResNet-50	1951	0.2778	0.2414	0.7281
EfficientNet-B0	1951	0.2742	0.2359	0.6520
MobileNetV3-Small	1951	0.2263 ± 0.0221	0.1873 ± 0.0154	0.5891 ± 0.0376

7 Calibration and Selective Prediction

[Draft: narrative pending completion of the three-seed matrix.]

8 Discussion

8.1 A Saturated Benchmark Carries Little Ranking Information

The three backbones separate cleanly on the laboratory test split, and the separations are statistically real: every pairwise comparison there is significant under Holm adjustment. But the spread that separates them is tiny in absolute terms, and it does not survive the domain shift. Under the shift, the backbone that places *last* in domain is statistically indistinguishable from the one that places *first* in domain ($p = 0.7488$ unadjusted, 1.0000 after Holm), while the ordering against MobileNetV3-Small reverses outright and is significant in both directions.

Table 4. Domain-shift degradation in macro-F1, mean $\pm$ sample standard deviation over three seeds. Drops are computed within each seed and then averaged, preserving the pairing between a checkpoint and its own cross-domain evaluation. The label spaces differ (38-way in domain, 21-way cross-domain), so this is a paired trend across models, not an isolated quantity. Accuracy-based drops are in the supplementary table.

Model	F1 in-dom.	F1 cross-dom.	Abs. drop	Rel. drop (\%)
ResNet-50	0.9899	0.2414	0.7485	75.61
EfficientNet-B0	0.9956	0.2359	0.7597	76.31
MobileNetV3-Small	0.9929 ± 0.0002	0.1873 ± 0.0154	0.8056 ± 0.0157	81.14 ± 1.56

Table 5. Expected calibration error, mean $\pm$ sample standard deviation over three seeds. ‘T’ is the temperature fitted on the held-out validation split and applied unchanged; it is never refitted on evaluation data. Temperature does not move the argmax, so accuracy is unchanged by it. Negative log-likelihood, Brier score, maximum calibration error, and AUC are in the supplementary table.

Model	Corpus	Setting	ECE	T
ResNet-50	PV	in-dom.	0.0903	1.0000
ResNet-50	PV	in-dom. (T)	0.0094	0.7086
ResNet-50	PV	cross-dom.	0.3038	1.0000
ResNet-50	PV	cross-dom. (T)	0.4305	0.7086
EfficientNet-B0	PV	in-dom.	0.0985	1.0000
EfficientNet-B0	PV	in-dom. (T)	0.0093	0.7030
EfficientNet-B0	PV	cross-dom.	0.2390	1.0000
EfficientNet-B0	PV	cross-dom. (T)	0.3778	0.7030
MobileNetV3-Small	PV	in-dom.	0.0987 ± 0.0036	1.0000 ± 0.0000
MobileNetV3-Small	PV	in-dom. (T)	0.0092 ± 0.0004	0.7002 ± 0.0019
MobileNetV3-Small	PV	cross-dom.	0.1478 ± 0.0172	1.0000 ± 0.0000
MobileNetV3-Small	PV	cross-dom. (T)	0.2900 ± 0.0157	0.7002 ± 0.0019
ResNet-50	PDC	in-dom.	0.1071	1.0000
ResNet-50	PDC	in-dom. (T)	0.0618	0.8350
EfficientNet-B0	PDC	in-dom.	0.1376 ± 0.0251	1.0000 ± 0.0000
EfficientNet-B0	PDC	in-dom. (T)	0.0564 ± 0.0111	0.7697 ± 0.0585
MobileNetV3-Small	PDC	in-dom.	0.1417 ± 0.0102	1.0000 ± 0.0000
MobileNetV3-Small	PDC	in-dom. (T)	0.1029 ± 0.0216	0.8427 ± 0.0673

That reversal is the substantive result, and it is worth stating precisely because the loose version of the claim is weaker than the true one. MobileNetV3-Small is significantly *better* than ResNet-50 on the laboratory test split and significantly *worse* than it under the shift. Both legs are significant after family-wise correction. This is not a re-ordering within noise; it is a genuine crossing.

The practical reading is uncomfortable for a literature organised around in-domain leaderboards. Choosing a backbone by its laboratory accuracy—the only signal available if one does not build a cross-domain evaluation—would have selected against the architecture that transfers best. A benchmark with no headroom cannot rank models on a property it does not measure, and by the time accuracies exceed 0.99, the remaining differences reflect fit to corpus-specific regularities at least as much as disease recognition [22,8].

8.2 In-Domain Calibration Does Not Transport

Temperature scaling [9] is close to free and close to universal advice: fit one scalar on held-out validation data, apply it, obtain better-calibrated confidences without touching accuracy. In domain, it works here exactly as advertised.

Across the domain shift it does not merely fail to help—it actively hurts, raising calibration error above the uncalibrated baseline for all three backbones. The mechanism is legible rather than mysterious. Label smoothing leaves the models mildly *under*-confident in domain, so the fitted temperature falls below 1 and *sharpens* the predictive distribution. Sharpening is precisely right in domain, where the model is usually correct. Out of domain the model is already overconfident relative to its much lower accuracy, and sharpening amplifies the error it was supposed to correct.

The deployment consequence inverts the usual recommendation: a temperature fitted on in-domain validation data should not be transported across a domain shift, and transporting it is worse than shipping the model uncalibrated. This is consistent with the broader finding that uncertainty quantification degrades under shift [17], but it is a sharper statement—the standard fix does not merely lose effectiveness, it changes sign. Because temperature does not move the argmax, a practitioner monitoring only accuracy would never see this happen. We report both variants side by side so the comparison is visible rather than implied.

8.3 Why the Controls Change What Can Be Concluded

Without the leakage audit, the honest form of our cross-domain result would have to be “degradation of at least this much, assuming no contamination.” With 0 exact overlaps established by exhaustive search and human adjudication of every near candidate, the assumption becomes a measurement.

Similarly, without a constrained label space, a reader could reasonably suspect the mapping had been tuned. Restricting it to a deterministic normaliser and a pre-existing policy decision means the shared space was not chosen to produce any particular number—and because it is conservative, its bias is in the

direction that makes our headline gap *larger* than a generous mapping would. We state that rather than leaving it for a reviewer to infer.

9 Limitations and Threats to Validity

1. **The two label spaces differ.** In-domain evaluation is 38-way; cross-domain is 21-way over a restricted space. The measured drop therefore combines domain shift with a change in task difficulty, and should be read as a paired trend across models rather than as an isolated quantity. The trend is the claim; the absolute magnitude is not.
2. **The shared mapping is conservative by construction.** Excluded near-misses are listed with reasons. A human-authored correspondence would likely enlarge the shared space and change the cross-domain numbers, most plausibly upward. Our numbers should be read as a lower bound on transfer under a mapping that refuses to guess.
3. **The field corpus carries documented label noise.** Images whose file-names contradict their assigned class are known to exist in PlantDoc. This bounds achievable accuracy on it and is not corrected here, because correcting it would alter the locked corpus and make the evaluation unreproducible against the published lock.
4. **The field test split is small.** With 225 images across 27 evaluable classes, per-class metrics have wide intervals; this is why we report bootstrap intervals rather than point estimates alone, and why per-class claims are avoided.
5. **One class cannot be evaluated.** The arthropod-pest class has two training images and no test images. It is neither learnable nor measurable and is retained only for corpus integrity.
6. **Statistical scope.** McNemar tests and bootstrap intervals are computed on seed 42. They characterise a fixed evaluation set for one trained checkpoint per backbone; they do not incorporate training-process variability, which is reported separately as across-seed spread. With three seeds, that spread is a descriptive quantity and we do not attach significance claims to it.
7. **Hardware.** Timing and throughput figures come from an Apple-silicon unified-memory accelerator and do not transfer to server GPUs in absolute terms.
8. **Cross-domain evaluation includes field-corpus training images.** All 1 951 shared-class images are used, since a laboratory-trained model has seen none of them. This is stated explicitly and the split composition is recorded in every result file.
9. **Governance status.** These results rest on a scientific experiment lock that binds digests, counts, taxonomy, and splits. It is not a completed formal dataset-governance freeze, and we do not claim otherwise.

10 Conclusion

We set out to measure how much of a laboratory-trained plant disease classifier’s accuracy survives contact with field imagery, under conditions where the

measurement can be trusted. Closing the two confounds that usually sit between such a comparison and its interpretation—train/test contamination, and a label correspondence chosen by the people reporting the result—turns out to change what can be concluded, not merely how carefully it must be hedged.

Two findings survive that scrutiny. First, a near-saturated in-domain benchmark carries almost no information about which backbone transfers: the in-domain ordering does not merely compress under the shift, it crosses, significantly and in both directions, between the largest and smallest backbones we tested. Selecting an architecture on laboratory accuracy is therefore not a conservative default—it is an active bet on a property the benchmark does not measure. Second, temperature scaling fitted on in-domain validation data degrades out-of-domain calibration below the uncalibrated baseline, because label smoothing drives the fitted temperature to sharpen a distribution that is already over-confident where it matters. A practitioner watching accuracy alone cannot observe this, since temperature never moves the argmax.

Neither finding requires a new architecture or a new corpus. Both required an evaluation built so that the alternative explanations could be ruled out rather than assumed away, and both point the same direction: for agricultural decision support, the evaluation protocol deserves at least as much scrutiny as the model. The full protocol, the dataset lock, the leakage audit, the mapping provenance, and the generation scripts for every table and figure in this paper are released with it.

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